



POLITECNICO
MILANO 1863

MPI

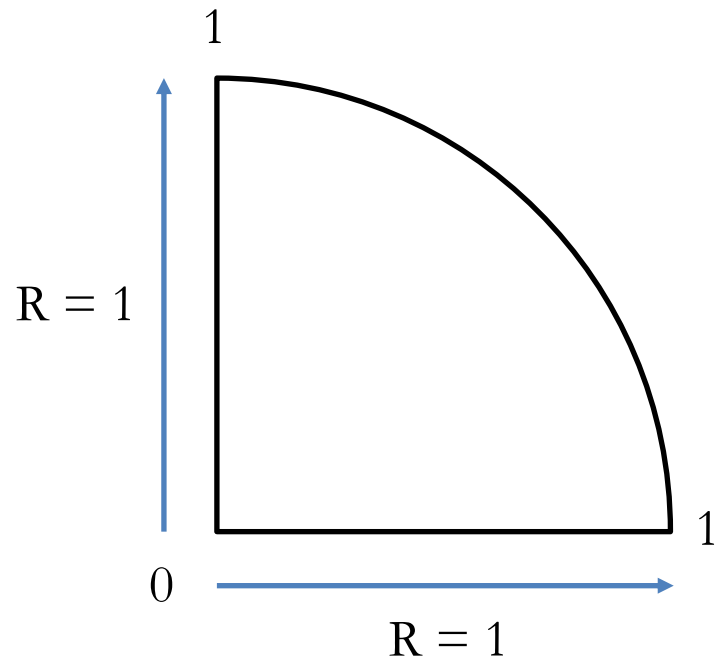
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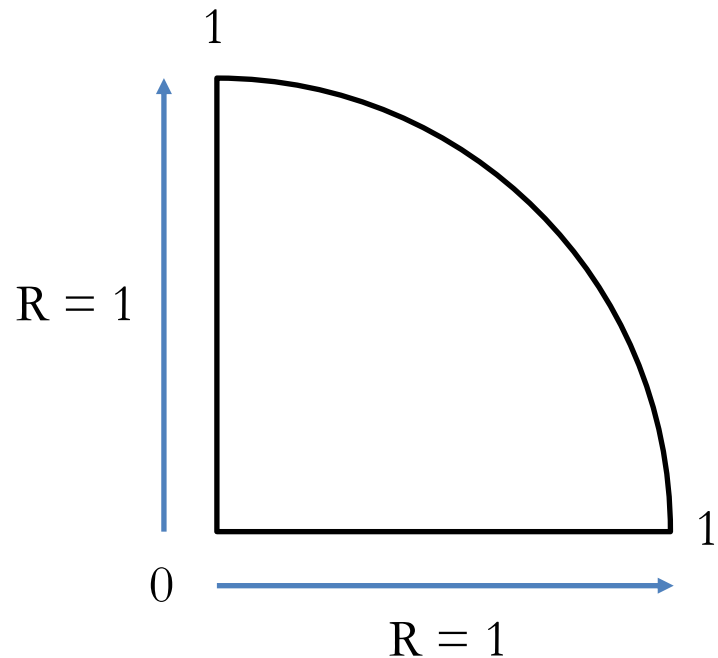
Exercise 1

- Use a Monte Carlo simulation to estimate the value of π
- Given a circle of radius $R=1$, its area is $A=\pi$
 - Let us consider only one fourth of the circle
 - Then the area is $A=\pi/4$



Exercise 1

- Consider the cartesian coordinates as in figure
- Given a point $P(x,y)$ it will be within the circle iff
 - $\text{Sqrt}(x^2+y^2) \leq \text{Sqrt}(1)$
 - $x^2+y^2 \leq 1$
- We can estimate A by generating many random points and checking how many of them fall within the circle
- We can then compute $\text{Pi} = 4*A$



Exercise 2

- Implement a guess game
 - N rounds
 - One process acts as leader and selects a number X between 1 and 1000 (included)
 - In the first round, the leader is process 0
 - All processes (including the leader) select a random number and send it to the leader
 - The process that selects the number that is closest to X wins the round and becomes the leader for the next round
 - If multiple processes have the same score, no one wins the round, and the leader does not change
 - Process 0 keeps track of the number of rounds won by each process and prints the updated leaderboard at the end of each round

Exercise 3

- You are to implement a simple traffic simulator
- Consider a linear road divided into consecutive segments numbered $0..N$ as in the figure below
 - N is represented by variable `num_segments` in the template
 - You may assume the number of segments to be a multiple of the number of processes



Exercise 3

- The simulation evolves in discrete rounds for a given number of iterations
 - `num_iterations` in the template file
- At each round
 - Some cars enter the road in segment 0
 - Use function `create_random_input()` to obtain the number of cars
 - Each car either remains in the same segment or moves to the next one
 - Use function `move_next_segment()` to determine if a car moves or not
 - Cars that move out of the last segment are not part of the simulation anymore

Exercise 3

- Every 10 iterations, you have to compute the total number of cars that are currently within the road
 - Process P0 prints the sum
 - The code for printing is already in the template
- Run the simulation in parallel on multiple processes, minimizing synchronization and communication as much as possible
- You may set `DEBUG` to 1 to obtain deterministic values and check the correctness of your code
 - At each round a single car enters the road
 - Cars always move to the next segment